



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	ELECTRICAL AND PHOTOVOLTAIC DEVICES				
Type of Course	Skill Enhancement Course 3				
Semester	VI				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	-	45
Pre-requisites	Basics of electromagnetism and electronics.				
Course Summary	This course explores the working of various electrical, photovoltaic and storage devices.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand and analyse the working of a DC motor.	U & An	P	Instructor-created exams / Home Assignments
CO2	Identify different electrical elements used in house wiring and demonstrate the working of these elements.	R & Ap	P	Instructor-created exams / Home Assignments

CO3	Explain various conventional and non-conventional power generation techniques and discuss the possibility of using these techniques in your state.	U	P	Instructor-created exams / Home Assignments
CO4	Analyse and determine the basic characteristics of Photovoltaic Cell. Design a model unit.	An & Ap	C	Instructor-created exams / Home Assignments
CO5	Explain the scope of different battery technologies and analyse the technical complexity to design the same.	Ap	P	Seminar Presentation / Group Tutorial Work
CO6	Generate skill to wind motors, wiring a home, develop storage devices.	Ap & C	P	Practical Assignment / Observation of Practical Skills / Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Mod ule	Unit	Content	Hrs (36 +9)	Mark s (50)
I	ELECTRICAL DEVICES		10	15
	1	DC motor basics: motor principle, comparison of generator and motor action, significance of the back e.m.f., voltage equation of a motor, condition of maximum power, torque, Armature torque of a motor, shaft torque, Speed of a DC motor, speed regulation, torque and speed of a DC motor, brief discussion on BLDC motor.	3	
	2	Working principle of a transformer, transformer construction, core-type transformers, shell type transformers.	1	
	3	Elementary theory of an ideal transformer, E.M.F. equation of a transformer, voltage transformation ratio, Transformer on load.	2	

	4	Classification of ac motors, induction motor: general principle, construction, Squirrel-cage rotor, phase-wound rotor.	2	
	5	Production of rotating field, three phase supply, mathematical proof, why does the rotor rotate.	2	
	Sections 29.1 – 29.11, 32.1 – 32.7, 32.10, 34.1 – 34.9 of chapter 29, 32 and 34 of Book 1			
II	BASICS OF WIRING, CONTROL AND SECURITY SYSTEMS		7	8
	6	Different types of wiring, Specifications of wires, types of cables.	2	
	7	Basics of wiring-Star and delta connection. Simple wiring schemes.	3	
	8	Fuses, Circuit breakers, earthing	2	
	Sections 11.1 – 11.3, 11.5 of chapter 11 of Book 2, sections 5.2, 5.3, 6.7- 6.9, 6.11 – 6.14 of chapter 5, chapter 6 of Book 3			
III	POWER GENERATION AND PHOTOVOLTAIC TECHNOLOGY		10	15
	9	Preference for electricity, comparison of sources of power, sources for generation of electricity, brief aspects of electrical energy systems, Conventional and non-conventional energy sources.	3	
	10	Photovoltaic materials: Introduction, Basic semiconductor physics	2	
	11	A generic photovoltaic cell, a more accurate equivalent circuit for a PV cell.	2	
	12	From cells to modules to arrays.	1	
	13	Crystalline silicon technologies	1	
	14	Thin film photovoltaic	1	
	Sections 24.1 – 24.6 of Book 1, sections 8.1 – 8.4, 8.8 (8.8.1- 8.8.2), 8.9 (8.9.1) of chapter 8 of Book 4			
IV	POWER STORAGE		9	12

	15	Introduction to energy sources	2	
	16	Battery technology: Lead acid batteries, Nickel metal hydride batteries.	2	
	17	Lithium batteries.	1	
	18	Voltage characteristics, standard and nomenclature, cell designs	1	
	19	Fuel cell types: types of fuel cells, complementary electrochemistry and thermodynamics of fuel cells (derivations not needed), solid oxide fuel cells, proton exchange membrane fuel cells.	3	
	Sections 1.1 – 1.8 of chapter 1, 4.1 – 4.4, 4.9 - 4.10 of chapter 4 and 8.1 – 8.3, 8.5(results in section 8.2 can be used, derivations not needed) of chapter 8 of Book 5.			
V	OPEN ENDED MODULE		9	
	1	Construction of a stepdown transformer a. 0-12 volt out b. 6-0-6 – 2 Amp out c. 800-Watt transformer for home ups		
	2	Rewind a household device motor (fan motor/mixer grinder motor/single phase water pump motor)		
	3	Create a miniature circuit including, isolator, rccb, mcb, single way switch and two-way switch.		
	4	Study the characteristics of photovoltaic cell		
	5	Construct a cylindrical capacitor by using aluminium foil and paper as dielectric.		
	6	Construct lead-acid cell.		
	7	Familiarise a battery management system (BMS) for a lithium-ion battery unit.		

Books and References:

1. A textbook of electrical technology by B. L. Thereja and A. K. Thereja, first multicolour edition (Book1)
2. Basic electrical engineering by C. L. Wadhwa, Fourth edition (Book 2)
3. Basic electrical engineering by Dr. K. Uma Rao and Dr. A. Jayalakshmi, revised edition 2014 (Book 3)
4. Wiring a house by Rex Cauldwell, , 4th edition, Publisher: The Taunton Press (Book 4)
5. Renewable and efficient electric power systems by Gilbert M Masters (Book 5)
6. Hydrogen, Batteries and Fuel Cells by Bengt Sunden Chapter1, 4, 8 (Book 6)

Mapping of COs with PSOs and POs :

	PS O1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	2	2	1	0	0	0	0	0	0	0	0
CO 2	2	1	2	2	1	0	0	0	0	0	0	0	0
CO 3	1	2	2	2	1	0	0	0	0	0	0	0	0
CO 4	2	1	1	2	2	1	0	0	0	0	0	0	0
CO 5	2	2	2	1	1	0	0	0	0	0	0	0	0
CO 6	2	1	2	2	1	0	0	0	0	0	0	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam

- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory /Practical Exam	Assignmen t /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	

MODEL QUESTION PAPERS
MAJOR CORE COURSES

Semester B.Sc. (CUFYUGP) Degree Examinations October 2024

PHY1CJ101: Fundamentals of Physics

(credits: 4)

Maximum Time: 2 hours

Maximum Marks: 70

Section A

[Answer All. Each question carries 3 marks] (Ceiling: 24 Marks)

1. If the two ends of a rope in equilibrium are pulled with forces of equal magnitude and opposite directions, why isn't the total tension in the rope zero?
2. When a car is hit from behind, the occupants may experience whiplash. Use Newton's laws of motion to explain what causes this result.
3. You drive a car up a steep hill at constant speed. Discuss all of the forces that act on the car. What pushes it up the hill?
4. A block rests on an inclined plane with enough friction to prevent it from sliding down. To start the block moving, is it easier to push it up the plane or down the plane? Why?
5. A rope tied to a body is pulled, causing the body to accelerate. But according to Newton's third law, the body pulls back on the rope with a force of equal magnitude and opposite direction. Is the total work done then zero? If so, how can the body's kinetic energy change? Explain.
6. Can the *total* work done on an object during a displacement be negative? Explain. If the total work is negative, can its magnitude be larger than the initial kinetic energy of the object?
7. If work W is required to stretch a spring a distance x from its unstretched length, what work (in terms of W) is required to stretch the spring an *additional* distance x ?
8. A projectile has the same initial kinetic energy no matter what the angle of projection. Why doesn't it rise to the same maximum height in each case?
9. Is it possible for a friction force to increase the mechanical energy of a system? If so, give examples.
10. A particle is in neutral equilibrium if the net force on it is zero and remains zero if the particle is displaced slightly in any direction. Sketch the potential energy function near a point of neutral equilibrium for the case of one dimensional motion.